$$|u\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \Longleftrightarrow \quad \sigma_z|u\rangle = |u\rangle$$
$$\sigma_x|u\rangle = |d\rangle$$
$$\sigma_y|u\rangle = i|d\rangle$$

$$|d\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \Longleftrightarrow \quad \sigma_z|d\rangle = -|d\rangle$$
$$\sigma_x|d\rangle = |u\rangle$$
$$\sigma_y|d\rangle = -i|u\rangle$$

$$|r\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \quad \Longleftrightarrow \quad \sigma_z|r\rangle = |l\rangle$$
$$\sigma_x|r\rangle = |r\rangle$$
$$\sigma_y|r\rangle = -i|l\rangle$$

$$|l\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{pmatrix} \quad \Longleftrightarrow \quad \sigma_z|l\rangle = |r\rangle$$
$$\sigma_x|l\rangle = -|l\rangle$$
$$\sigma_y|l\rangle = i|r\rangle$$

$$|i\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix} \quad \Longleftrightarrow \quad \sigma_z|i\rangle = |o\rangle$$
$$\sigma_x|i\rangle = i|o\rangle$$
$$\sigma_y|i\rangle = |i\rangle$$

$$|o\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} \end{pmatrix} \quad \Longleftrightarrow \quad \sigma_z|o\rangle = |i\rangle$$
$$\sigma_x|o\rangle = -i|i\rangle$$
$$\sigma_y|o\rangle = -|o\rangle$$